

Khoshrav Doctor

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RESEARCH SUMMARY

Ph.D. candidate in Robotics specializing in multi-agent systems, learning-based control, and autonomous model learning. Experienced in ROS-based robotic systems, multi-agent coordination, planning and reinforcement learning for decision-making under uncertainty. Proven track record of publications and deploying algorithms on real robotic platforms.

EDUCATION

University of Massachusetts Amherst

Ph.D. in Computer Science (GPA 3.95)

Laboratory for Perceptual Robotics

Expected Graduation: May 2026

University of Massachusetts Amherst

Master of Science in Computer Science (GPA 3.94)

Laboratory for Perceptual Robotics

Relevant Courses: Robotics, Reinforcement Learning, Machine Learning, Artificial Intelligence, Feedback Control Systems

Graduated: Feb 2020

K. J. Somaiya College of Engineering, Mumbai

Bachelor of Engineering in Information Technology with First Class (GPA 7.1 out of 10)

Graduated: July 2016

WORK EXPERIENCE

Laboratory for Perceptual Robotics, UMass Amherst

Aug 2017 – Present

Graduate Research Assistant

- Developing learning-based exploration and model-learning methods for partially observable robotic systems, improving sample efficiency by up to 4x and enabling more effective belief space planning under uncertainty
- Investigating methods to learn reusable skills hierarchically to speed up grasping and transport of objects
- Implementing control systems for a 3 degree of freedom robotic head with an RGB-D camera
- Performing routine diagnosis, maintenance and repair in the uBot robotic platform
- Technologies used: ROS (C++ and Python development), Gazebo, Deep Learning, Reinforcement Learning

Indiana University

Feb 2016 – Present

Research Affiliate

- Analyzing IMU movement data to develop computational, non-invasive and objective approaches for detecting Autism Spectrum Disorder, Attention-Deficit/Hyperactivity Disorder and their comorbidity
- Designing Deep Learning and Machine Learning algorithms based on systems neuroscience to achieve 71% accuracy using LSTM networks
- Publishing and presenting relevant work at numerous physics and neuroscience meetings

TRAC Labs, Inc.

Sept – Dec 2021, May – Aug 2022, June – Aug 2023

Graduate Robotics Intern

- Developed multi-agent coordination systems to improve teleoperator situational awareness by leveraging additional heterogeneous robotic agents
- Improved multi-agent path planning algorithms for real-time robotic systems
- Built visualization and debugging tools (RViz) to analyze planning behavior, system performance, and failure modes
- Contributed towards experimental design and writing for research papers, reports, and documentation of projects

Platform for Ethical and Responsible Computing Education (PEaRCE)

Feb 2025 – Dec 2025

Head Teaching Assistant (TA)

- Directed two six-person teams to embed C.S. ethical education in courses serving 1100 students
- Supervised creation of software tools that automated TA administrative processes, increasing capacity from 200 to 600 students per semester
- Coordinated deploying ethical simulations between students, course instructors and PEaRCE research staff
- Facilitated ethics-centered discussions for 5 courses, helping students critically assess the societal and human impacts of emerging technologies they develop
- Identified and implemented improvements to PEaRCE website to enhance user experience and feedback using React and Django

Software Developer

- Designed and implemented online video analysis platform Newton for Search, Summarization and Statistics
- Developed and maintained aiSentinel Security Suite features like Camera Tampering Detection and Asset Tracking
- Researched technologies such as CMT for Object Tracking
- Interacted directly with potential clients to identify and implement their requirements into our products
- Technologies used: C++, NodeJS, Python, PHP, Caffe, QT

SELECTED PUBLICATIONS

Deep Learning Diagnosis Plus Kinematic Severity Assessment of Neurodivergent Disorders | Scientific Reports, 2025

Learning Intrinsically Motivated Transition Models for Autonomous Systems | IEEE International Conference on Development and Learning, 2022

Evaluating Sensorimotor Abstraction on Curricular for Learning Manipulation Skills | IEEE International Conference on Development and Learning, 2022

Learning to Manipulate in Open Environments | Workshop on Learning of Manual Skills in Humans and Robots, 2020

Learning From Less Data: A Unified Data Subset Selection and Active Learning Framework for Computer Vision | Winter Conference on Applications of Computer Vision, 2019

Demystifying Multi-Faceted Video Summarization: Tradeoff Between Diversity, Representation, Coverage and Importance | Published at Winter Conference on Applications of Computer Vision, 2019

RELEVANT ACADEMIC PROJECTS

Spatial Model Building

- Leveraged deep learning mechanisms and a 3 Degree of Freedom robotic head to build probabilistic models representing the location of objects in free space
- Used learned models to sample actions to solve an object localization task in dynamic environments

Chess Automated: An automated physical chess board

- Implemented a website using JavaScript and chessboard.js for a remote player
- Prototyped online game play between multiple boards using Processing and MyRobotLab
- Developed an Android application which runs a chess engine and communicates moves to the chess board
- Contributed Arduino code that controls the pieces' movement using an X-Y plotter mechanism

TECHNICAL SKILLS

- Robotics and AI: ROS, Multi-agent Systems, Planning, Reinforcement Learning, Deep Learning, SolidWorks
- Programming Languages: C, C++, Python, Java, MATLAB, Arduino, Processing
- Web Development: JavaScript, HTML, CSS, PHP

TEACHING ASSISTANTSHIPS

- COMPSCI 220 (Fall 2017): Programming Methodology
- COMPSCI 240 (Spring 2018): Reasoning Under Uncertainty
- COMPSCI 503 (Fall 2018): Embedded Systems Engineering
- COMPSCI 603 (Spring 2020): Robotics
- COMPSCI 187 (Fall 2022): Programming with Data Structures
- COMPSCI 683 (Spring 2023): Artificial Intelligence
- COMPSCI 589 (Spring 2024): Machine Learning
- Platform for Ethical and Responsible Computing (Fall 2024, Spring 2025, Fall 2025)

AWARDS

- Dissertation Writing Fellowship, Summer 2024, MCICS, UMass Amherst
- Proposal Writing Fellowship, Fall 2023, MCICS, UMass Amherst
- 1st Place Project Competition, Tatva Convergence 2016
- 2nd Place Project Competition, Prakalpa 2016
- Highest Committed Student Branch Activist, 2014, Computer Society of India
- Jewel of Somaiya Award for Outstanding Performance, 2013, Somaiya Education Trust